VisionGuard

Catching vehicle theft & break-ins the moment they happen — on the cheap, dark cameras societies already own.

Concept note & prototype status · Founder: [your name] · [email / phone] · [FILL: date]

The problem

The world has close to a billion CCTV cameras, and almost all of them are **passive** — they record for review *after* a crime, not prevention *during* one. In residential societies, dozens of cameras run that nobody watches. When a car is broken into at night, the guard finds out the next morning, from the footage. The camera saw everything and did nothing.

Existing "AI cameras" don't fix this because they are built and demoed on clean daylight video. Real CCTV is **low-resolution, dark, and low-frame-rate** — exactly where they fail. And generic motion alerts are so noisy that people switch them off. A noisy security system is worse than none.

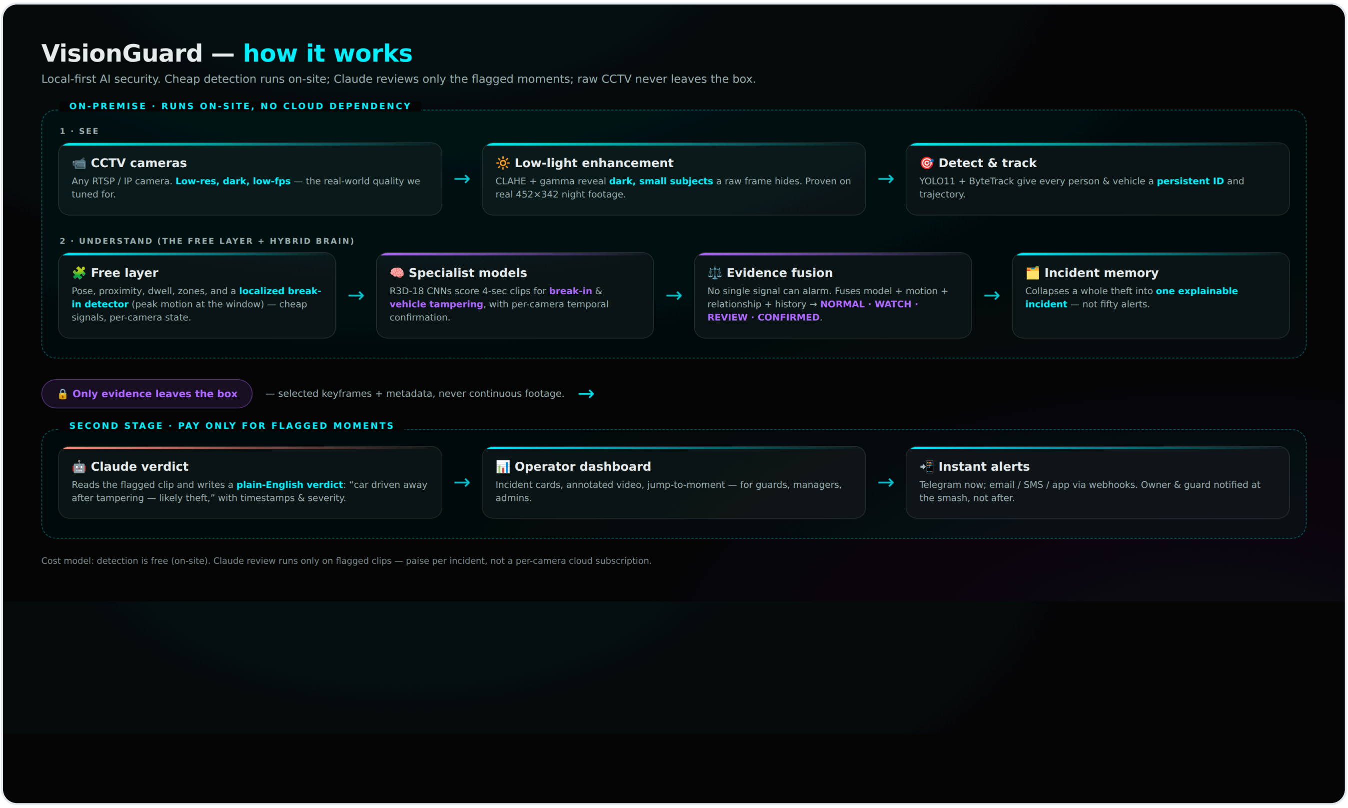
The solution

VisionGuard turns a society's existing cameras into a watchful AI that:

- **Sees in the dark** — enhances low-light frames so it detects the person a raw feed hides.
- **Understands the act** — fuses a person at a vehicle, a localized break-in motion, and the car leaving into a single *theft* conclusion.
- **Explains itself** — Claude reviews the flagged clip and writes a plain-English verdict with timestamps and severity.
- **Alerts instantly** — owner and guard are notified at the smash, not the next morning.

How it works

Local-first: cheap detection runs on-site; only flagged evidence goes to Claude for a verdict; **raw CCTV never leaves the box**.



The VisionGuard pipeline: enhance → detect & track → free-layer signals + specialist models → evidence fusion → incident memory → Claude verdict → dashboard & alerts.

It works on real, bad footage — the evidence

We diagnosed and fixed detection on **actual 452×342 night-CCTV theft footage** (not a lab clip). Two measured results from that real video:

Same frame, before and after our low-light enhancement. In the raw feed the intruder at the car window is black-on-black and invisible to the detector; enhanced, the person and window are clearly visible.

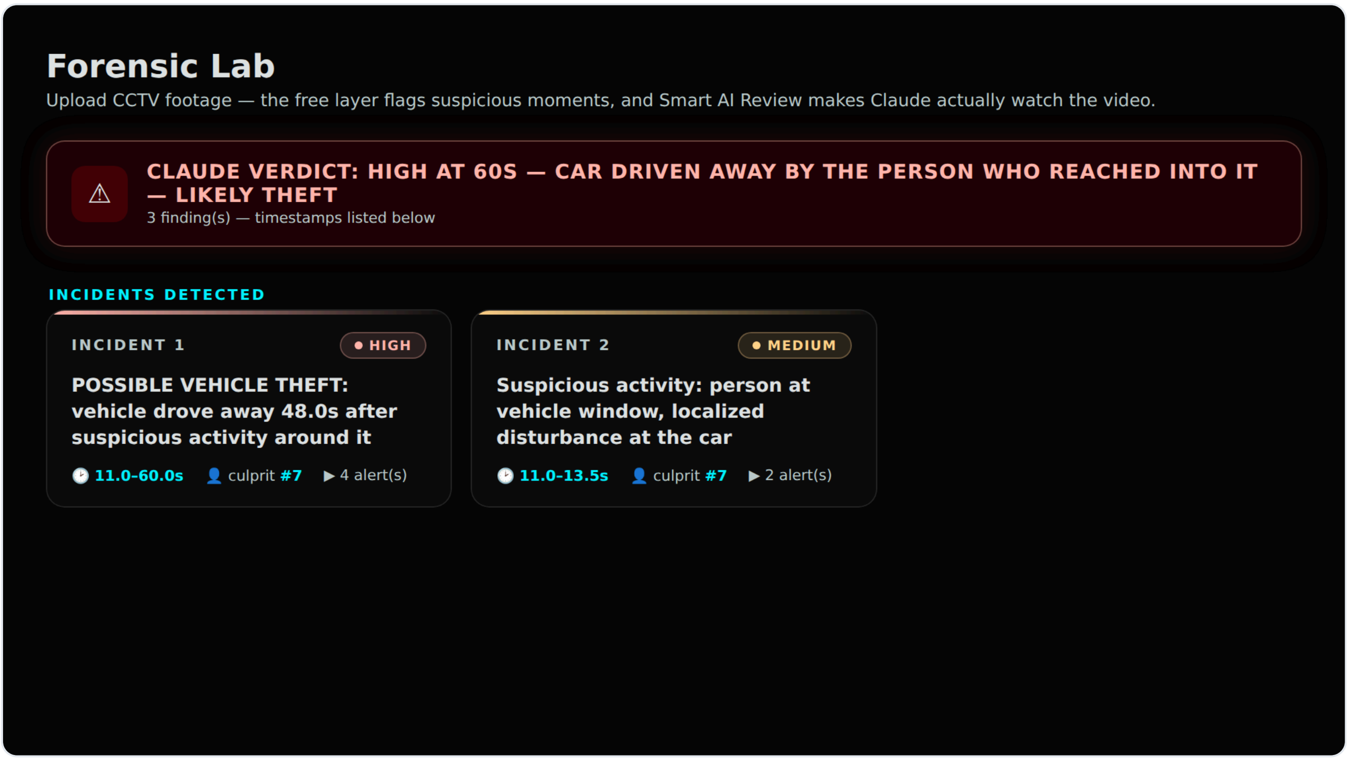
101 → 142
FRAME BRIGHTNESS AFTER ENHANCEMENT (PERSON BECOMES DETECTABLE)

90–160
LOCALIZED BREAK-IN SIGNAL DURING THE ACT (VS ~15 WHEN QUIET)

180
AUTOMATED TESTS PASSING ACROSS THE PIPELINE

The break-in leaves a clean, well-separated signal at this exact quality — but only when measured as **localized** motion at the contact point, which we built. The whole system then collapses the event into one explainable incident and Claude writes the verdict.

What an operator sees



The analysis view: Claude's plain-English verdict on top, then the theft as a single incident card (time span, culprit ID, severity) — not a wall of alerts. Click a card to jump the video to that moment.

Where the prototype is in the build

VisionGuard is a **working, end-to-end prototype**, validated qualitatively on real footage. It is honest about what is proven and what the next stage delivers.

AREA	STATUS	DETAIL
Detection on real low-res night CCTV	**Built ✓**	Low-light enhancement + YOLO/ByteTrack tracking; person & vehicle detection proven on the actual clip.
Break-in / theft understanding	**Built ✓**	Localized motion detector + free-layer signals + theft-chain, measured on real footage.
Specialist models + fusion	**Built ✓**	R3D-18 CNNs (break-in, vehicle tampering) with per-camera temporal confirmation; evidence fusion so no single signal can raise a false alarm.
Claude verdict & incident memory	**Built ✓**	Plain-English verdict per incident; per-frame alerts merged into one incident.
Operator dashboard + Telegram alerts	**Built ✓**	Upload-and-analyze "Forensic Lab", incident cards, annotated playback, instant alerts.
Published accuracy number	**In progress** ⏳	Requires an untouched, labeled multi-camera holdout — the deliberate gap this raise closes.
Live multi-camera & mobile app	**Next →**	Live pipeline, roles/permissions, mobile alerts, email/SMS routing.
Model hardening on domain data	**Next →**	Train on our own society footage + surveillance datasets (SPHAR, VIRAT) to cut domain-shift.

Tech: Python · FastAPI · YOLO11 + ByteTrack · R3D-18 (PyTorch) · OpenCV · SQLite · Claude (Anthropic) for scene-verdict review · Telegram alerts. Local-first architecture; footage stays on-site.

Why now, and where we win

Why now: cameras are everywhere and AI is finally good enough to watch them live — but nobody has made it work on the cheap, dark cameras people actually own. We have. **The market** (AI video surveillance) is already several billion dollars and growing fast.

Our white space vs. well-funded incumbents (Verkada, Ambient.ai globally; Staqu, Wobot, Awiros in India): most target enterprise/retail/city or sell their own expensive hardware. We serve **residential societies on their existing cheap cameras**, tuned for **low-quality footage**, with **explainable, local-first** alerts at a **paise-per-incident** cost. The moat is our performance on bad footage plus the data flywheel from real deployments.

The ask

Raising [FILL: ₹ amount] to reach [FILL: milestone — e.g. validated recall + first paying societies]. Use of funds: data collection + GPU to harden the models, pilot deployments, and the product build (live multi-camera, mobile, front-end).